

INVESTMENT PORTFOLIO ANALYSIS USING SQL, PYTHON AND DASHBOARD VISUALIZATION

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CHAPTER ONE: INTRODUCTION

1.1 Background

Investment portfolio analysis is an important process in financial management that involves evaluating the performance, profitability, and risk exposure of investment assets within a portfolio. Investors allocate capital across different financial instruments such as stocks, bonds, cryptocurrencies, money market funds, savings instruments, and land investments in order to maximize returns while minimizing financial risk.

Traditionally, portfolio management relied heavily on manual calculations and intuition-based decision-making. However, technological advancements and the growth of financial data analytics have transformed portfolio analysis into a highly data-driven discipline. Modern investment management now depends on analytical tools capable of processing large datasets efficiently and generating accurate financial insights.

SQL databases are widely used for storing and querying financial data efficiently, while Python programming supports statistical analysis, automation, and visualization development. Dashboard visualization tools further improve investment monitoring by presenting portfolio insights through charts, KPI cards, and interactive reporting systems.

Portfolio analysis helps investors:

- Evaluate investment performance
- Measure portfolio returns
- Assess volatility and risk exposure
- Analyze portfolio diversification
- Identify profitable investment opportunities
- Improve allocation strategies

The integration of SQL, Python, and dashboard visualization tools improves analytical accuracy and enhances evidence-based investment decision-making.

This project focused on analyzing an investment portfolio using SQL, Python, and dashboard visualization tools to evaluate portfolio performance, allocation, and risk-adjusted returns.

1.2 Problem Justification

The increasing complexity of financial markets has made investment portfolio management more challenging for both individual investors and organizations. Modern portfolios often contain multiple assets with varying return characteristics and risk levels, making manual analysis inefficient and unreliable.

Traditional portfolio management methods are often time-consuming and prone to calculation errors. Investors may struggle to identify underperforming assets, monitor portfolio volatility,

and evaluate diversification effectively. Poor portfolio monitoring may lead to concentration risk, inefficient capital allocation, and reduced financial performance.

Analytical technologies such as SQL, Python, and dashboard visualization tools provide practical solutions for addressing these challenges. SQL improves data storage and querying efficiency, Python supports statistical analysis and automation, while dashboard tools simplify interpretation of financial insights.

This study is important because it demonstrates how analytical technologies can improve portfolio monitoring, support data-driven investment decisions, and enhance financial performance evaluation.

1.3 Problem Statement

Investment portfolios consist of multiple financial assets with varying levels of profitability, volatility, and market risk. Effective portfolio management requires accurate evaluation of returns, portfolio allocation, diversification, and risk-adjusted performance. However, many investors lack integrated analytical systems capable of processing and interpreting investment data efficiently.

Manual portfolio analysis techniques are insufficient for handling modern investment datasets because they reduce analytical accuracy and limit the ability to identify portfolio trends and risk exposure. Consequently, investors may make poor financial decisions due to limited analytical insights.

Although SQL, Python, and dashboard visualization tools are widely used in financial analytics, there is limited practical integration of these technologies within a single portfolio analysis framework. Therefore, this study seeks to develop an integrated analytical system capable of cleaning investment data, calculating portfolio returns, measuring risk exposure, and visualizing financial performance effectively.

1.4 OBJECTIVES

1.4.1 Broad Objective

To analyze an investment portfolio using SQL, Python, and dashboard visualization tools in order to evaluate portfolio performance, allocation, and risk-adjusted returns.

1.4.2 Specific Objectives

- To evaluate portfolio risk using volatility analysis.
- To identify the best-performing and worst-performing assets.
- To generate recommendations for improving portfolio management.

1.5 HYPOTHESIS

There is a significant relationship between the use of analytical technologies and the effectiveness of portfolio performance evaluation and investment decision-making.

1.6 RESEARCH QUESTIONS

- How can SQL support portfolio data management and analysis?
- How can Python improve portfolio return calculations and risk evaluation?
- What insights can dashboard visualization tools provide regarding portfolio performance?

1.7 SCOPE AND LIMITATIONS

The project focused on analyzing a portfolio investment dataset containing multiple investment categories including:

- Stocks
- Cryptocurrencies
- Fixed-income investments
- Money market funds
- Savings instruments
- Land investments

The analysis involved:

- Data cleaning
- SQL querying
- Return calculations
- Portfolio allocation analysis
- Risk analysis
- Dashboard visualization

Due to limited project duration, the study relied on historical portfolio data stored in CSV format instead of live financial market data.

CHAPTER TWO: LITERATURE REVIEW

2.1 Investment Portfolio Analysis

2.1.1 Overview of Investment Portfolios

An investment portfolio refers to a collection of financial assets held by an investor for the purpose of generating income, preserving wealth, and achieving financial growth. Portfolio management focuses on selecting appropriate investments and balancing risk and return.

According to Markowitz (1952), diversification is essential for minimizing portfolio risk while maximizing expected returns. Diversification reduces dependence on individual asset performance by distributing investments across multiple financial instruments.

2.1.2 Portfolio Diversification

Portfolio diversification involves spreading investment capital across different asset classes to reduce exposure to market volatility. Diversified portfolios generally exhibit lower overall risk because losses from one investment may be offset by gains from another.

Research indicates that effective diversification improves portfolio stability and long-term financial sustainability.

2.1.3 Risk and Return Analysis

Risk and return are key concepts in financial analysis. Investors generally expect higher returns from investments associated with higher risk levels.

Portfolio risk is commonly measured using statistical indicators such as variance and standard deviation. Standard deviation measures volatility by evaluating how far investment returns deviate from the average return.

The Sharpe Ratio is widely used for evaluating risk-adjusted performance because it measures returns relative to portfolio volatility.

2.1.4 SQL in Financial Analytics

SQL is widely used in financial analytics because it supports efficient storage, querying, and retrieval of structured financial data. Analysts use SQL to:

- Retrieve portfolio records
- Calculate returns
- Aggregate investment categories
- Generate financial summaries
- Rank investment performance

SQL improves analytical efficiency and supports integration with dashboard systems.

2.1.5 Python in Portfolio Analysis

Python has become one of the most widely used programming languages in financial analytics because of its flexibility and powerful analytical libraries.

Libraries such as:

- pandas
- numpy
- matplotlib

Support:

- Data cleaning
- Statistical analysis
- Financial calculations
- Visualization generation

Python enables analysts to process large investment datasets accurately and efficiently.

2.1.6 Dashboard Visualization in Financial Reporting

Dashboard visualization tools improve interpretation of portfolio insights through:

- KPI cards
- Pie charts
- Scatter plots
- Bar charts
- Trend graphs

Dashboards simplify financial reporting and improve evidence-based investment decision-making.

2.1.7 Gaps in Literature

Although portfolio management and financial analytics have been widely studied, there remains limited practical integration of SQL, Python, and dashboard visualization tools within a single analytical workflow.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology used to conduct the investment portfolio analysis project.

3.2 Research Design

The project adopted a quantitative analytical research design because it involved statistical analysis of numerical investment data.

The research process involved:

- Data collection
- Data cleaning
- SQL querying
- Python analysis
- Dashboard visualization

3.3 Dataset Description

The project used a CSV dataset named investment_data.csv containing investment portfolio records.

Dataset Variables

Variable	Description
Asset	Name of investment asset
Type	Investment category
Amount	Amount invested
Initial Price	Purchase price
Current Price	Current market price
Risk Adjusted Score	Total return relative to the risk taken
Total Return	Calculated return
Allocation	Portfolio allocation percentage

3.4 Data Cleaning

The dataset contained formatting inconsistencies including:

- Currency symbols
- Percentage signs
- Missing values
- Inconsistent column names

Python preprocessing techniques were used to clean and standardize the dataset.

3.5 SQL Analysis

SQL was used to:

- Create database tables
- Query investment records
- Calculate returns
- Aggregate investment categories
- Rank investment performance

CHAPTER FOUR: SQL RESULTS AND ANALYSIS

4.1 SQL Query Outputs

The SQL analysis generated outputs related to:

- Portfolio returns
- Type allocations
- Asset rankings

Sum of investment amount query

The screenshot displays the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo, the project name 'My Project 9482', and a search bar. Below this, the 'Sandbox' section provides a link to upgrade to the full BigQuery experience. The main workspace is divided into three panes: the Explorer on the left, the query editor in the center, and the query results on the right.

The Explorer pane on the left shows a sidebar with navigation options: Home, Starred, Shared with me, Job history, and a dropdown for the current project 'sustained-flux-491218-p7'. Under this project, there are links to Datasets, Connections, Conversations, Agents, Queries, (Classic) Queries, Notebooks, and Data canvases.

The central query editor pane shows a query titled 'Untitled query' with the following SQL code:

```
1 SELECT SUM(Amount)
2 FROM `sustained-flux-491218-p7.Investment_analysis_data.investment_analysis_project` LIMIT 1000
```

Buttons for 'Run', 'Download', 'Share', 'Schedule', 'Open in', 'More', and 'Save' are located above the query editor. The 'Run' button has been clicked, resulting in a 'Query completed' notification and a 'Using on-demand processing quota' message.

The right pane displays the 'Query results' section, which includes tabs for 'Job information', 'Results', 'Visualization', 'JSON', 'Execution details', and 'Execution graph'. The 'Results' tab is active, showing a table with one row and one column:

Row	f0_
1	1000000

Below the table, there are buttons for 'Chat', 'Save results', and 'Open in'. At the bottom right of the results pane, there is a message: 'Activate Windows Go to Settings to activate Windows.' The bottom status bar shows 'Results per page: 50' and '1 - 1 of 1'.

Investment amount by type query

The screenshot shows the Google Cloud BigQuery console interface. The URL bar indicates the project is 'sustained-flux-491218-p7'. The left sidebar shows the project structure with 'sustained-flux-491218-p7' selected. The main panel displays a query titled 'Untitled query' with the following SQL code:

```
1 SELECT type, SUM(Amount)
2 FROM `sustained-flux-491218-p7.Investment_analysis_data.investment_analysis_project`
3 GROUP BY type;
```

The query has been executed successfully, and the results are displayed in a table. The table has columns 'Row', 'type', and 'Amount'.

Row	type	Amount
1	Fixed income fund	100000
2	Money market fund	132000
3	Special fund	250000
4	Real estate investment trust	60000

The interface also shows a 'Query completed' message and a 'Using on-demand processing quota' notification. The bottom status bar indicates the time is 21:22 on 05/05/2026.

Investment amount in descending order query

The screenshot shows the Google Cloud BigQuery console interface. The URL bar indicates the project is 'sustained-flux-491218-p7'. The left sidebar shows the project structure with 'sustained-flux-491218-p7' selected. The main panel displays a query titled 'Untitled query' with the following SQL code:

```
1 SELECT asset, amount
2 FROM `sustained-flux-491218-p7.Investment_analysis_data.investment_analysis_project`
3 ORDER BY amount DESC
4 LIMIT 8;
```

The query has been executed successfully, and the results are displayed in a table. The table has columns 'Row', 'asset', and 'amount'.

Row	asset	amount
1	Mansa X	250000
2	Kajiado	200000
3	EUR CIO ACC Growth fund	152000
4	Kuza USD	132000

The interface also shows a 'Query completed' message and a 'Using on-demand processing quota' notification. The bottom status bar indicates the time is 21:27 on 05/05/2026.

Risk adjusted score in descending order query

The screenshot shows the Google Cloud BigQuery console interface. The query editor displays the following SQL query:

```
1 SELECT 'Risk adjusted score'
2 FROM
3   `sustained-flux-491218-p7.Investment_analysis_data.investment_analysis_project`
4 ORDER BY 'Risk adjusted score' DESC;
5
```

The query has been executed successfully, and the results are displayed in a table. The table has two columns: 'Risk adjusted score' and an unnamed column. The results are sorted in descending order.

Row	Risk adjusted score
1	0.32
2	0.12
3	0.11
4	0.11

The interface also shows the Explorer panel on the left with the project 'sustained-flux-491218-p7' selected. The bottom status bar indicates the time is 21:39 on 05/05/2026.

Asset return calculations query

The screenshot shows the Google Cloud BigQuery console interface. The query editor displays the following SQL query:

```
1 SELECT
2   Asset,
3   Type,
4   (
5     SAFE_CAST(REGEXP_REPLACE('Current Price', r'[^\d-\.]*', '') AS FLOAT64)
6     - SAFE_CAST(REGEXP_REPLACE('Initial Price', r'[^\d-\.]*', '') AS FLOAT64)
7   ) / NULLIF(
8     SAFE_CAST(REGEXP_REPLACE('Current Price', r'[^\d-\.]*', '') AS FLOAT64)
9   ) AS return
10
```

The query has been executed successfully, and the results are displayed in a table. The table has four columns: 'Asset', 'Type', and 'return'. The results are sorted by 'Asset'.

Row	Asset	Type	return
1	Zimele	Fixed income fund	0.105
2	Kuza USD	Money market fund	0.045454545454...
3	Mansa X	Special fund	0.192

The interface also shows the Explorer panel on the left with the project 'sustained-flux-491218-p7' selected. The bottom status bar indicates the time is 19:56 on 17/05/2026.

Asset with the highest return query

The screenshot shows the Google Cloud BigQuery console interface. The Explorer on the left lists the project 'sustained-flux-491218-p7'. The main query editor displays the following SQL query:

```
1 SELECT
2   Asset,
3   AVG(
4     (
5       SAFE_CAST(
6         REGEXP_REPLACE('Current Price', r'[^\d-9.]', '')
7         AS FLOAT64
8       )
9     )
10  ) AS avg_return
```

The query has been executed successfully, as indicated by the 'Query completed' status. The results are displayed in a table with the following data:

Row	Asset	avg_return
1	Safaricom	0.883280757097...

The interface also shows a 'Query results' section with tabs for Job information, Results, Visualization, JSON, Execution details, and Execution graph. The 'Results' tab is currently selected. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 20:24 on 17/05/2026.

Asset with the lowest return query

The screenshot shows the Google Cloud BigQuery console interface. The Explorer on the left lists the project 'sustained-flux-491218-p7'. The main query editor displays the following SQL query:

```
1 SELECT
2   Asset,
3   AVG(
4     SAFE_DIVIDE(
5       current_price - initial_price,
6       initial_price
7     )
8   ) AS avg_return
```

The query has been executed successfully, as indicated by the 'Query completed' status. The results are displayed in a table with the following data:

Row	Asset	avg_return
1	Acorn I-REIT	0.036665211697...

The interface also shows a 'Query results' section with tabs for Job information, Results, Visualization, JSON, Execution details, and Execution graph. The 'Results' tab is currently selected. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 20:34 on 17/05/2026.

The SQL outputs showed differences in investment allocation and highlighted high-performing and underperforming assets within the portfolio.

CHAPTER FIVE: PYTHON ANALYSIS

5.1 Purpose of Python

Python was used for:

- Data preprocessing
- Return calculations
- Risk analysis
- Statistical analysis
- Visualization generation

5.2 Return Calculations

Portfolio returns were calculated using:

$$\text{Return} = \{\text{Current Price}\} - \{\text{Initial Price}\} / \{\text{Initial Price}\}$$

Python implementation:

```
df["return"] =  
    (df["current_price"] - df["initial_price"])  
    / df["initial_price"]
```

5.3 Portfolio Allocation

Portfolio allocation was calculated using:

$$\text{Allocation} = \{\text{Asset Amount}\} / \{\text{Total Portfolio Investment}\}$$

5.4 Risk Analysis

Portfolio risk was measured using standard deviation:

Python implementation:

```
risk = df["return"].std()
```

5.5 Sharpe Ratio Analysis

The Sharpe Ratio was calculated using:

$$\text{Sharpe Ratio} = \{\text{Average Return}\} / \{\text{Risk}\}$$

Python implementation:

```
sharpe = avg_return / risk
```

5.6 Python Outputs

The screenshot shows the PyCharm IDE with a Python file named `main.py` open. The Run console displays the output of a script. The first 5 rows of the dataset are shown, followed by the dataset shape before cleaning.

```
C:\Users\Wughanga\PycharmProjects\HelloWorld\investment_analysis_project\.venv\Scripts\python.exe C:\Users\Wughanga\PycharmProjects\HelloWorld\investment_...

FIRST 5 ROWS
asset_data ... risk_(low/med/high)
0 Vasili Africa ... Low (1)
1 Vasili Africa ... Low (1)
2 Mansa X Factsheet ... Medium (2)
3 portal.vuka.co.ke ... Medium (2)
4 www.sc.com ... Medium (2)

[5 rows x 13 columns]

DATASET SHAPE BEFORE CLEANING
(15, 13)

MISSING VALUES
asset_data 7
asset 7
type 6
country 7
initial_price 7
current_price 7
amount 7
%allocation 7
annual_return_% 7
```

106:1 (104 chars, 3 line breaks) CRLF UTF-8 4 spaces Python 3.12 (investment_analysis_project)

The screenshot shows the PyCharm IDE with the same Python file `main.py`. The Run console displays the output of a script, showing the dataset shape after cleaning and the cleaned numeric columns.

```
%allocation 7
annual_return_% 7
dividend_yield_% 7
total_returns_% 7
risk_adjusted_score 7
risk_(low/med/high) 7
dtype: int64

DATASET SHAPE AFTER CLEANING
(8, 13)

CLEANED NUMERIC COLUMNS
amount current_price initial_price
0 100000.0 110500.00 100000.00
1 132000.0 138000.00 132000.00
2 250000.0 298000.00 250000.00
3 60000.0 23.75 22.91
4 152000.0 131.75 107.37
5 46000.0 2358.00 1738.00
6 60000.0 29.85 15.85
7 200000.0 212000.00 200000.00

TOTAL INVESTED
1000000.0
```

106:1 (104 chars, 3 line breaks) CRLF UTF-8 4 spaces Python 3.12 (investment_analysis_project)

```
investment_analysis_project  Version control  main  Run  main  x  18:07 17/05/2026
```

```
ALLOCATION
```

	asset	allocation
0	Zimele	0.100
1	Kuza USD	0.132
2	Mansa X	0.250
3	Acorn I-REIT	0.060
4	EUR CIO ACC Growth fund	0.152
5	Ethereum	0.046
6	Safaricom	0.060
7	Kajiado	0.200

```
CATEGORY BREAKDOWN
```

type	
Crypto currency	40000.0
Exchange traded fund	152000.0
Fixed income fund	100000.0
Land	200000.0
Money market fund	132000.0
Real estate investment trust	60000.0
Special fund	250000.0
Stock	60000.0

```
Name: amount, dtype: float64
```

Activate Windows
Go to Settings to activate Windows.

investment_analysis_project > main.py 106:1 (104 chars, 3 line breaks) CRLF UTF-8 4 spaces Python 3.12 (investment_analysis_project)

```
investment_analysis_project  Version control  main  Run  main  x  18:08 17/05/2026
```

```
PORTFOLIO RISK
```

```
0.28276742766462615
```

```
AVERAGE RETURN
```

```
0.23827470977813353
```

```
SHARPE RATIO
```

```
0.8426526058748789
```

```
PORTFOLIO RETURN
```

```
0.1826203482256743
```

```
BEST PERFORMING ASSET
```

asset	return
Safaricom	0.083281

```
Name: 6, dtype: object
```

```
WORST PERFORMING ASSET
```

asset	return
Acorn I-REIT	0.036665

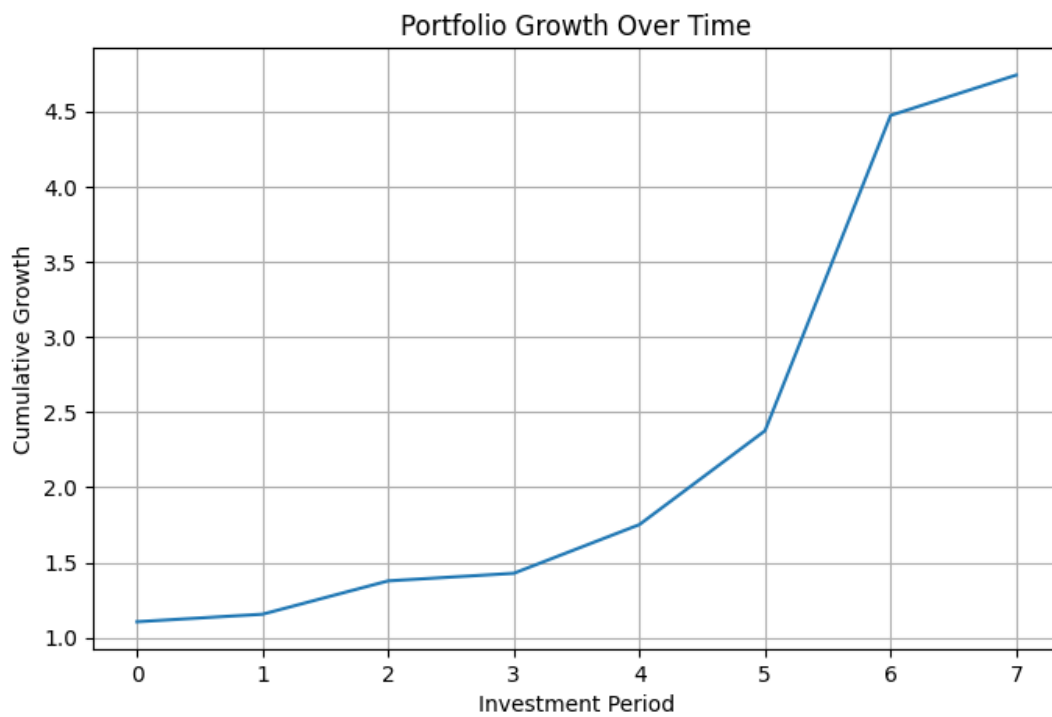
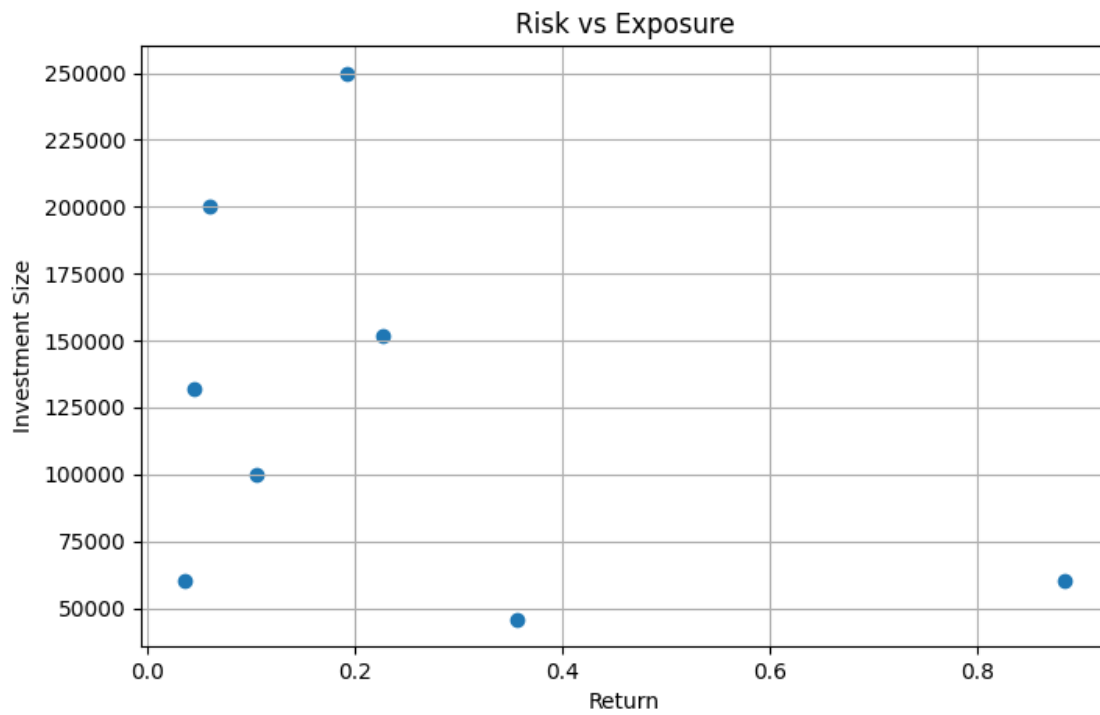
```
Name: 3, dtype: object
```

```
ANALYSIS COMPLETE
```

```
Results saved to analysis_results.csv
```

Activate Windows
Go to Settings to activate Windows.

investment_analysis_project > main.py 106:1 (104 chars, 3 line breaks) CRLF UTF-8 4 spaces Python 3.12 (investment_analysis_project)



The Python outputs demonstrated portfolio return trends, allocation patterns, and portfolio volatility analysis.

CHAPTER SIX: DASHBOARD VISUALIZATION

6.1 Dashboard Purpose

The dashboard consolidated portfolio metrics into interactive visual reports for easier interpretation and decision-making.

6.2 Dashboard Components

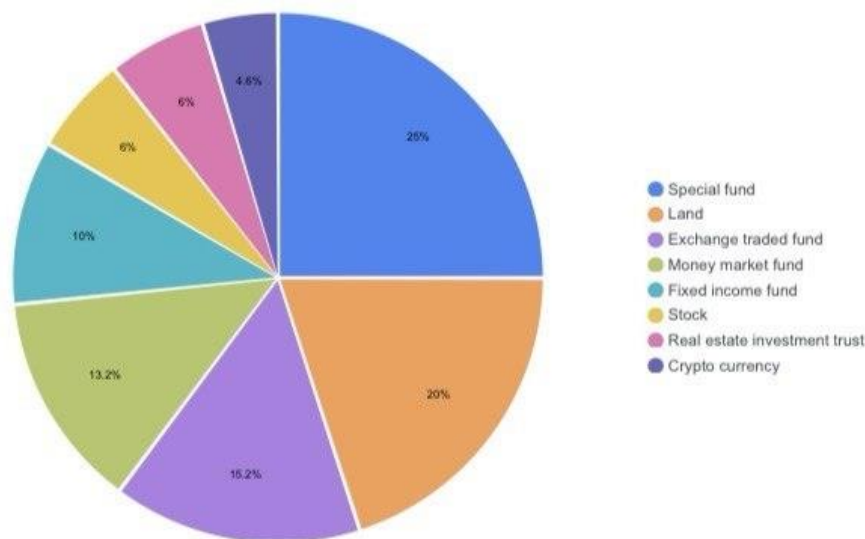
The dashboard included:

- KPI cards
- Pie charts
- Scatter plots
- Bar charts
- Performance rankings

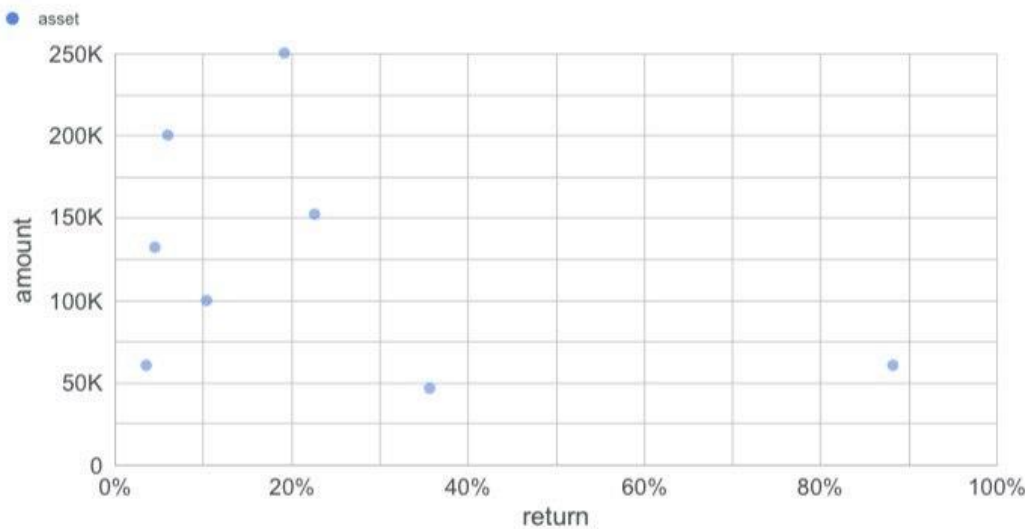
Investment Analysis Portfolio Dashboard

Total invested	Average Return	Portfolio Risk	Sharpe Ratio
amount	return	return	risk_adjusted_score
1,000,000	23.83%	28.28%	0.11

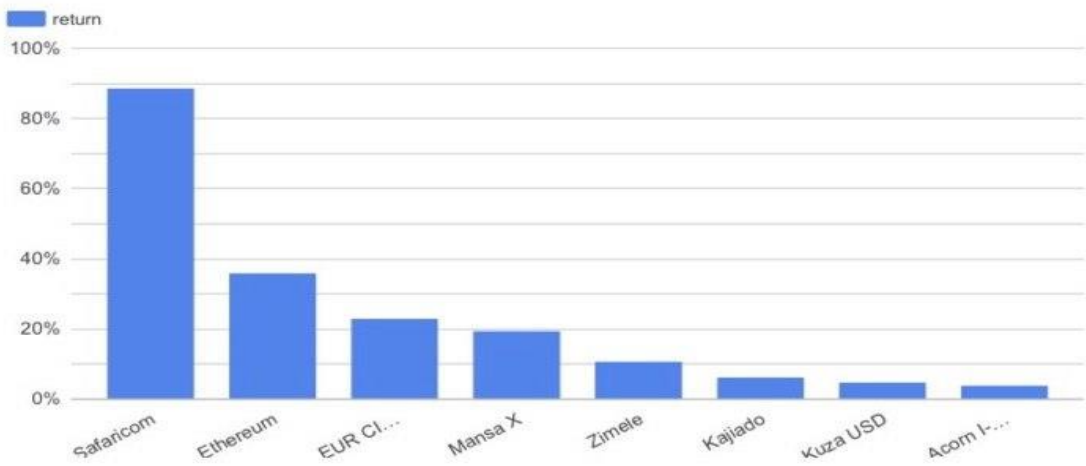
Portfolio Allocation



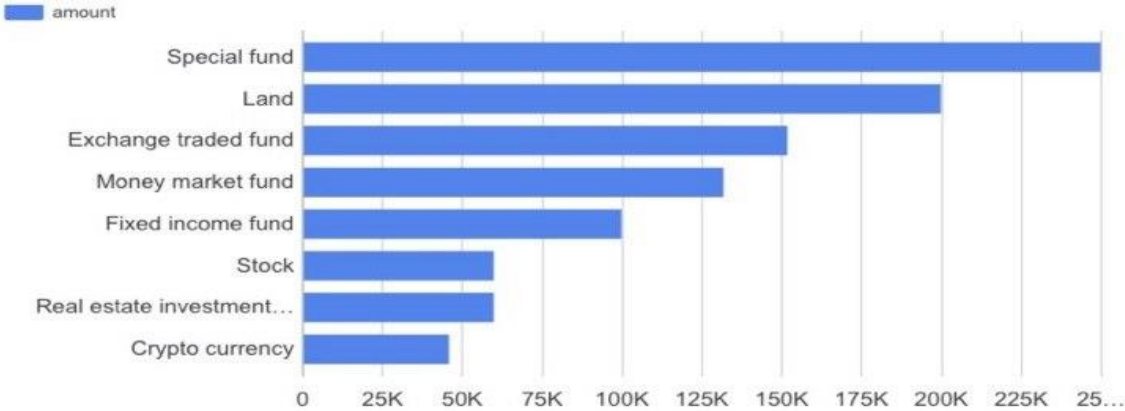
Return vs Investment Exposure



Asset Performance Comparison



Investment by Type



Asset Performance Ranking

	asset	type	return ▾	amount
1.	Safaricom	Stock	88.33%	60,000
2.	Ethereum	Crypto currency	35.67%	46,000
3.	EUR CIO ACC Growth fund	Exchange traded fund	22.71%	152,000
4.	Mansa X	Special fund	19.2%	250,000
5.	Zimele	Fixed income fund	10.5%	100,000
6.	Kajlado	Land	6%	200,000
7.	Kuza USD	Money market fund	4.55%	132,000
8.	Acorn I-REIT	Real estate investment trust	3.67%	60,000

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The dashboard simplified portfolio interpretation and improved communication of analytical findings.

CHAPTER SEVEN: RESULTS AND DISCUSSION

7.1 Portfolio Diversification

The portfolio demonstrated diversification across multiple investment categories, reducing overall portfolio risk exposure.

7.2 High-Performing Assets

Certain assets generated significantly higher returns than others, contributing substantially to portfolio profitability.

7.3 Portfolio Concentration

Some investment categories accounted for a large percentage of total invested capital, increasing concentration risk.

7.4 Risk Exposure

High-return investments also exhibited higher volatility, demonstrating the relationship between risk and return.

7.5 Risk-Adjusted Performance

The Sharpe Ratio analysis showed that the portfolio generated moderate returns relative to its level of risk exposure.

7.6 Importance of Data Cleaning

The project highlighted the importance of preprocessing investment data before analysis. Data cleaning improved analytical accuracy and reliability.

7.7 Data Limitations

Total return calculations exclude dividends for Safaricom and Acorn I-REIT. These assets account for 12% of portfolio weight, resulting in an estimated impact of <2% on total portfolio returns.

CHAPTER EIGHT: CONCLUSION AND RECOMMENDATIONS

8.1 Conclusion

This project successfully demonstrated the integration of SQL, Python, and dashboard visualization tools in investment portfolio analysis.

The project involved:

- Data cleaning
- SQL querying
- Return calculations
- Risk analysis
- Dashboard visualization
- Performance ranking

SQL improved structured data management while Python supported advanced statistical analysis and visualization generation. Dashboard systems enhanced communication of portfolio insights through interactive charts and KPI indicators.

The study demonstrated that analytical technologies significantly improve investment monitoring and evidence-based portfolio management.

8.2 Recommendations

Improve Diversification

Investors should diversify investments across additional low-correlation asset classes to reduce portfolio risk exposure.

Reduce Portfolio Concentration

Portfolio allocation should be adjusted to avoid excessive concentration in specific investment categories.

Monitor Volatile Assets

Highly volatile investments should be monitored continuously to manage downside risk effectively.

Increase Stable Investments

Increasing exposure to fixed-income and money market instruments may improve portfolio stability.

Adopt Automated Analytics

Organizations and investors should adopt SQL, Python, and dashboard systems for continuous portfolio monitoring and analytical reporting.

CHAPTER NINE: REFERENCES

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